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Introduction to Design Vision

Learning Objectives

After completing this lab, you should be able to:

- Read a design and related files into Design Vision, the graphical interface to **Design For Test Compiler (DFTC)**
- Compile that design into a test-ready design
- Explore that design in Design Vision
- Save the test-ready design



Lab Duration:
30 minutes

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Overview

All files for this lab except for designs and libraries are located in the directory `lab5_gui`.

Directory Structure

<code>lab5_gui</code>	Current working directory
<code>analyzed</code>	Intermediate <code>acs_read_hdl</code> files
<code>logs</code>	Session log files
<code>unmapped</code>	Unmapped protocol
<code>mapped</code>	Gate-level netlist
<code>mapped_scan</code>	Will contain final netlist
<code>reports</code>	DFTC reports
<code>tmax</code>	Files for downstream tools
<code>scripts</code>	Constraint and run scripts
<code>../ref/rtl/RISC_CORE/vhdl</code>	Design files
<code>../ref/db</code>	Library db files

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question, consult the back portion of this lab for help.

Instructions

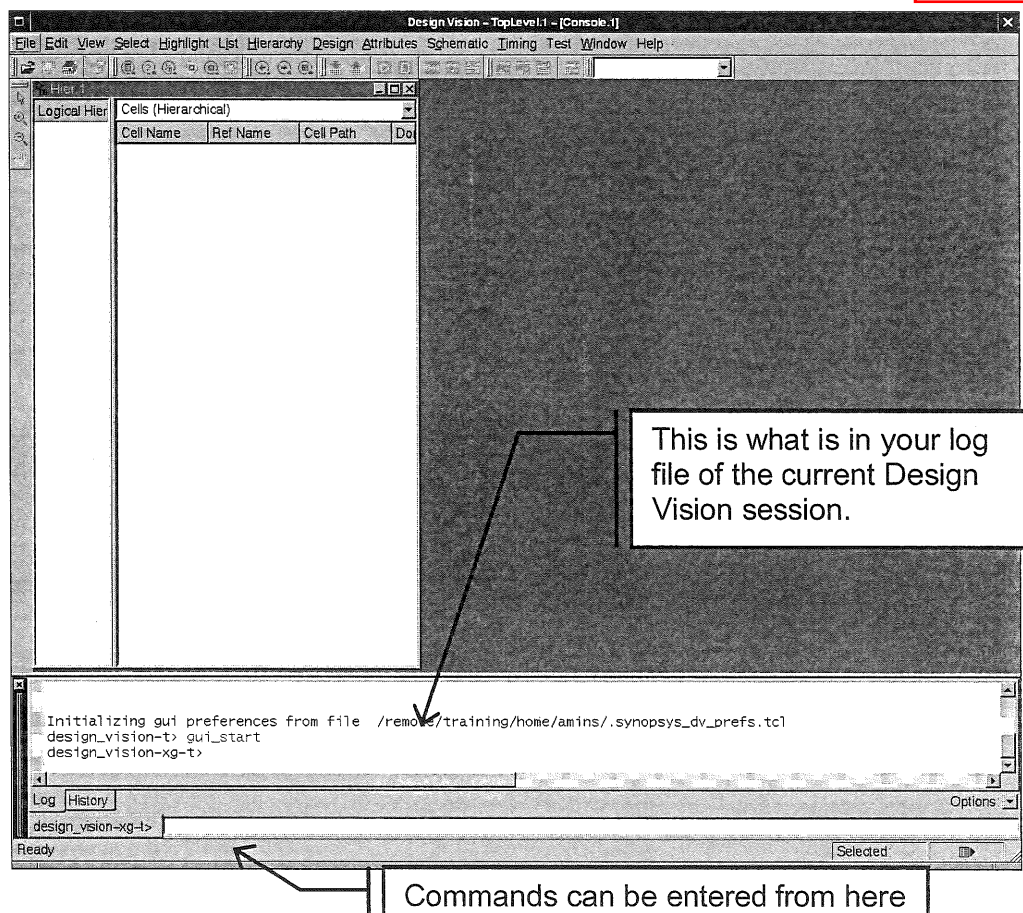
Task 1. Read a design into Design Vision

1. Invoke Design Vision.

```
UNIX% cd lab5_gui
UNIX% design_vision | tee logs/gui.log
```

Question 1. What is the difference between **Design Vision** and **dc_shell**?

> design_vision
=
> dc_shell -gui
图形化用户界面
与
终端命令行模式



Once in Design Vision, you can take advantage of its graphical interface for debugging and troubleshooting problems, or work in `dc_shell`, taking advantage of the speed of a shell.

2. Read the mapped RISC_CORE design, by applying the following command:

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① read_ddc 是否不再
需要电脑上 .db 库

read_ddc mapped/RISC_CORE.ddc

Task 2. Explore the Design

When inserting scan into your design, problems may occur. Being able to explore your design graphically is a powerful debugging aid.

1. Resize your Session Window, making it smaller so that you can focus on the graphical display at the top portion of the GUI.

The following figure shows some key points for navigating in Design Vision.

Display a schematic. Click on the **orange AND gate** to see a schematic of the design. To look at a schematic for a particular sub-block, select that sub-block in the Design Overview window and then click on the orange gate.

Current Design. In this case, RISC_CORE is the current design.

Design Overview Window. Once the designs have been read into Design Vision, this window shows the top-level block and all of the subblock names. A + sign indicates subblocks exist beneath. For instance, select the + sign next to I_STACK_TOP to see the subblocks of the STACK_TOP block.

Transcript Window: Here it shows the results of the link command. Many commands return 1 or 0 to indicate success or failure, respectively. Be aware that other commands return collections or other values.

Use the Options pulldown to search the log for Errors, Warnings, Informational messages, or commands. You can also use F3.

Command history log

Cell Name	Ref Name	Cell Path
I_ALU	ALU	I_ALU
I_CONTROL	CONTROL	I_CONTROL
I_DIV_CLK	DIV_CLK	I_DIV_CLK
I_DATA_PATH	DATA_PATH	I_DATA_PATH
I_INSTRN_LAT	INSTRN_LAT	I_INSTRN_LAT
I_PRGRM_CNT_TOP	PRGRM_CNT_TOP	I_PRGRM_CNT_TOP
I_REG_FILE	REG_FILE	I_REG_FILE
I_RST	RST	I_RST

Question 2. What is the current design?

```
design_vision> current_design
Current design is 'RISC_CORE'.
(RISC_CORE)
```

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2. Display the current design from the TCL shell.

You can also display the current design from the TCL shell by using the `current_design` command. Try it:

```
current_design
```

3. Display all the designs from the TCL shell by using the `list_designs` command.

```
list_designs
```

```
design_vision> current_design
Current design is 'RISC_CORE'.
(RISC_CORE)
design_vision> list_designs
CONTROL          PRGRM_CNT      REG_FILE      STACK_MEM_0
DATA_PATH        PRGRM_CNT_TOP  RISC_CORE (*) STACK_MEM_1
DIV_CLK          PRGRM_DECODE   RST           STACK_MEM_2
INSTRN_LAT       PRGRM_FSM      STACK_FSM      STACK_TOP
```

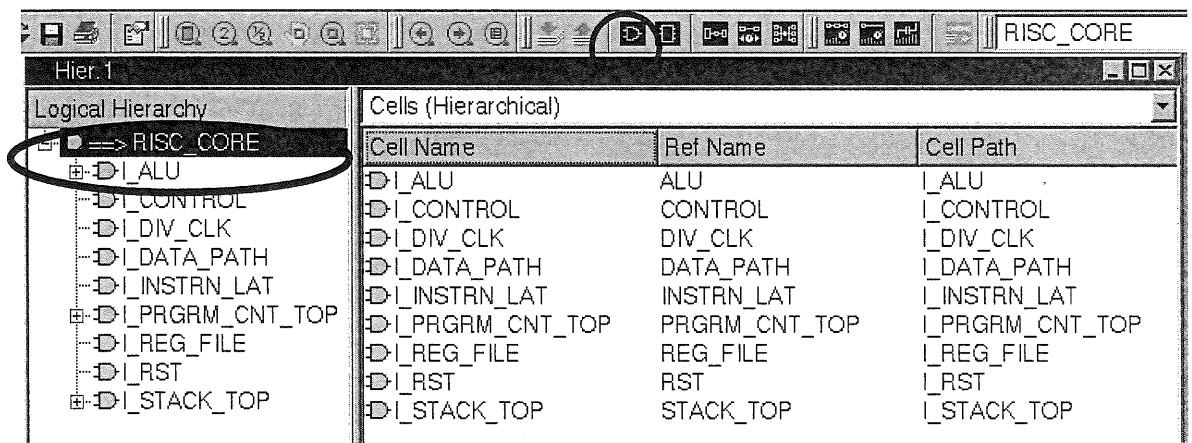
4. Display all the designs graphically.

Select the pull down arrow to the right of the current design; the results match those from the `list_designs` command.

5. Display a schematic of the entire RISC_CORE ASIC.

Specify which part of the ASIC you want to see.

In this case, select RISC_CORE from the Design Overview Window. Notice a list of the subdesigns below RISC_CORE is displayed



Notice the Cell name and ref name of each subdesign are shown. Cells names are also called **instances**. Ref names are also called **designs**.

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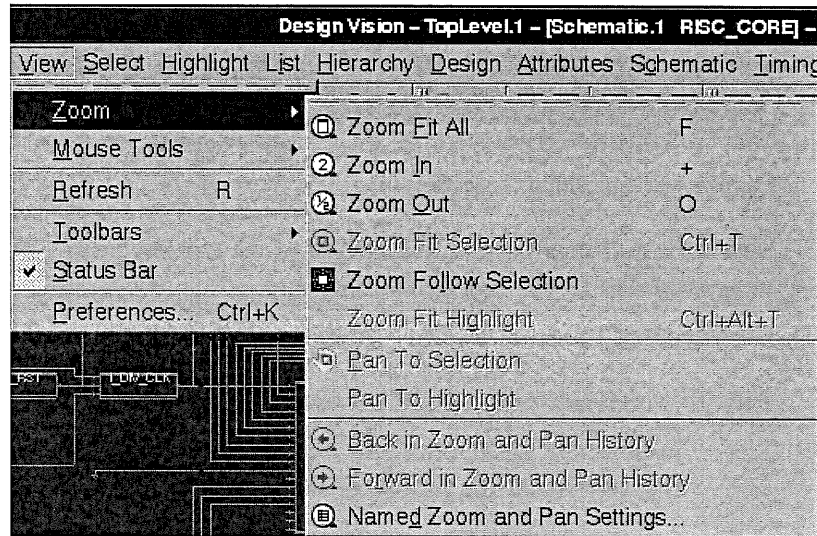
6. Display a schematic of the RISC_CORE ASIC.

Select the orange AND gate in the toolbar. You should see a schematic of the RISC_CORE ASIC.

You can zoom in and out to see things more clearly.

7. Zoom in to examine the schematic in more detail.

Select the **view** pull-down menu to zoom in and out



Note: You may first have to select an area of the schematic which has nothing in it with the mouse to be able to use the Zoom features.

Once in Zoom In mode, you may need to click on the Selection Tool before changing to another mode, such as Zoom Out. The escape key (ESC) is the shortcut for the Selection Tool.

You can also use the middle mouse button to zoom and pan, using so called strokes, e.g. depress the middle mouse button then move the mouse from SW to NE. You should have zoomed in.

8. Display the symbol view (input and output ports) of the RISC_CORE ASIC.

Select the green IC chip icon next to the orange AND gate.



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9. Verify that your entire design is mapped to the gates.

The **report_hierarchy** command shows you the entire hierarchy of the design in a textual format, as well as the cells in the design and libraries to which they belong. In this case, you are using the **tsmc** library. If any cells belong to the **gtech** library, you can conclude that there are unmapped cells in your design.

Run the command:

```
report_hierarchy
```

10. Exit Design Vision

```
exit
```

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看 schematic. 看输入变量 signal.

Task 3. Debug Violations using the GUI

1. Invoke Design Compiler in 64 bit mode

```
UNIX% dc_shell -64 | tee logs/debug.log
```

2. Read the mapped design

```
read_ddc mapped/RISC_CORE.ddc
```

3. Run a design rule check.

```
dft_drc
```

Question 3. What error message do you get?

..... No model found on design

```
dc_shell> dft_drc
Error: No model found on design
0
dc_shell>
```

4. Resolve DFTC's complaint; that is, create a test protocol.
(?) 未 set-dft-signal.

```
create_test_protocol
```

5. Run a design rule check again.

```
dft_drc
```

multi-clock protocol. requires that the strobe time be before a clock's pulse if it is used for transition fault testing.

Question 4. Is the design free of DFT problems or does the report contain many violations?

Start the Design Vision GUI

```
start_gui
```

```
DRC Report
Total violations: 356
-----
5 TOPOLOGY VIOLATIONS
5 Unconnected input pin violations (TEST-332)
-----
351 PRE-DFT VIOLATIONS
261 Uncontrollable clock input of flip-flop violations (D1)
2 DFF set/reset line not controlled violations (D2)
88 DFF set/reset line not controlled violations (D3)
-----
Warning: Violations occurred during test design rule checking. (TEST-124)
-----
Sequential Cell Report
261 out of 261 sequential cells have violations
-----
SEQUENTIAL CELLS WITH VIOLATIONS
+ 261 cells have test design rule violations
```

6. Bring up the violation browser by using the menu **Test → Browse Violations**.
7. Re-run **dft_drc** to populate the violation browser **Test → Run DFT DRC**.

Note: If you run **dft_drc** from the Design Vision GUI command prompt, the violation browser will automatically be launched.

8. Click on the “+” in front of “PRE-DFT”, then click on the D1 category.

② Violation Browser 中 pre-DFT 什么意思
TOPOLOGY

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① D1/D2/D3 是如何分类 clode / input / reset?

Select the first violation on the right. On the bottom, you will see an explanation of the error.

9. Right click on the first D1 violation, then select “Inspect Violation”, or click on the “Inspect Violation” button on the lower right. A schematic should appear.

Question 5. What logic value is present on the clock?

clk 上标为 x. ☒

Question 6. What DFTC command is needed to specify this test clock?

`set_dft_signal -view exist -type ScanClock -timing {45 55} -port Clk`

10. Type the command that you just noted.

```
set_dft_signal ...
```

11. Recreate the test protocol, then switch back to the Violation Browser, and click on “Run DFT DRC”. The schematic window should disappear.

Note: The test protocol is recreated by simply re-running the `create_test_protocol` command.

`remove_test_protocol`

Question 7. Was the number of violations affected at all?

```
DRC Report
Total violations: 353

.....

TOPOLOGY VIOLATIONS
5 Unconnected input pin violations (TEST-332)

PRE-DFT VIOLATIONS
250 Uncontrollable clock input of flip-flop violations (D1)
2 DFF set/reset line not controlled violations (D2)
88 DFF set/reset line not controlled violations (D3)

Warning: Violations occurred during test design rule checking. (TEST-124)

Sequential Cell Report
261 out of 261 sequential cells have violations

.....

SEQUENTIAL CELLS WITH VIOLATIONS
* 261 cells have test design rule violations
```

12. Once again, “Inspect Violation” for the first D1 violation shown.

Question 8. What value is shown for TEST_MODE?

☒

Question 9. What DFTC command is needed to specify this test mode?

`set_dft_signal -view exist -type Constant -active 1 -port TEST_MODE`

13. Type the command that you just noted.

```
set_dft_signal ...
```

14. Recreate the test protocol, then switch back to the Violation Browser, and click on “Run DFT DRC”.

Question 10. Was the number of violations affected at all?

15. Analyze the first D3 violation.

```
DRC Report
Total violations: 95

.....

TOPOLOGY VIOLATIONS
5 Unconnected input pin violations (TEST-332)

PRE-DFT VIOLATIONS
2 DFF set/reset line not controlled violations (D2)
88 DFF set/reset line not controlled violations (D3)

Warning: Violations occurred during test design rule checking. (TEST-124)

Sequential Cell Report
90 out of 261 sequential cells have violations

.....

SEQUENTIAL CELLS WITH VIOLATIONS
* 90 cells have test design rule violations
SEQUENTIAL CELLS WITHOUT VIOLATIONS
* 171 cells are valid scan cells
* 18 cells are non-scan shift-register cells
```

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Question 11. What value is shown for Reset?

X

Question 12. What DFTC command is needed to specify this asynchronous reset (the reset is active low)?

`set_dft_signal -view exist -type Reset -active 0 -port Reset`

16. Type the command you just noted.

```
set_dft_signal ...
```

17. Recreate the test protocol, then switch back to the Violation Browser, and click on "Run DFT DRC".

Question 13. Was the number of violations affected at all?

18. Exit Design Vision.

Congratulations! You have completed this lab.

```
-----
DRC Report
-----
Total violations: 5
-----
5 TOPOLOGY VIOLATIONS
  5 Unconnected input pin violations (TEST-332)
-----
Warning: Violations occurred during test design rule checking. (TEST-124)
-----
Sequential Cell Report
-----
0 out of 261 sequential cells have violations
-----
SEQUENTIAL CELLS WITHOUT VIOLATIONS
+ 243 cells are valid scan cells
+ 18 cells are non-scan shift-register cells
-----
```